Wellness Studio OS — Project Execution, Delivery & Implementation Plan

Document 6 of 6 · Version 1.0 · 29 September 2026 Sources: Master PRD, Solution Architecture, Module Specification, Database Design, API Contract. Scope: execution strategy only; nothing from Documents 1–5 is regenerated.

Source conflicts the team must settle on Day 1 (owner: Solution Lead):

#	Conflict	Plan decision
1	PRD §16 puts AI in the Phase 1 "must complete" list. Module Spec §10.3 lists the AI assistants as "P1 – within first sprint", after the P0 MVP.	AI is built Day 4–5 but is **cut first** (PRD cut order). Classified P1 below.
2	Module Spec says 19 ERP DocTypes and 8 n8n workflows. The Database Design says 20 DocTypes; the Module Spec §8 lists 9 workflows.	Build to the **Database Design** (20) and the Module Spec §8 list (9, including Event Reminder as P2).
3	Module Spec: Redis in the stack, API p95 < 500 ms. Architecture/PRD: Redis optional, booking p95 < 1 s.	Redis only for OTP/rate-limit if needed. Test target **< 1 s booking, < 500 ms simple reads**.
4	The sources name only four of the six team members.	Architecture §14.7 role proposal adopted; members are referred to by role code (SL, DBE, AIP, AIQ, ERPE, FEL).
5	Architecture: one shared Beta/Staging environment doubles as the demo environment; Production comes in Phase 2.	"Deployment" in this plan means **Dev + one Beta/Demo environment**.

1. Project Delivery Overview

Project goal. Build Wellness Studio OS, a multi-tenant, India-first, WhatsApp-native studio operating system with ERPNext as the system of record. AI acts only through governed APIs.

MVP goal. In 5 days, demonstrate the canonical flow end to end on a deployed Beta environment, with one pilot studio on the multi-tenant model: Member → Membership → Class → Booking → Attendance → Payment → Notification → Dashboard. This runs alongside a narrow Admin AI Assistant (two read queries and Create Member).

Success criteria (demo-day gate)

#	Criterion	Source
1	Canonical flow passes E2E on Beta	PRD §16
2	50 parallel last-seat requests produce exactly one confirmation	NFR-PRF-03, F-03
3	Cross-tenant access is denied and audit-logged	F-06, R-09
4	Payment is Paid only after a verified Razorpay signature; duplicates create no second record	F-05, DI-03
5	Every P0 event produces one Notification Log record in mock mode, with no duplicates	F-07
6	Attendance correction writes an immutable audit record	F-08
7	Assistant answers match dashboard figures; no false success; no execution without confirmation	AI-01
8	Zero Sev-1 defects; no secrets in Git; the demo runs from the deployed environment	PRD DoD

Key assumptions

- A-01 to A-10 of the PRD hold. Razorpay and WhatsApp are in test/mock mode, and production credentials are not needed.
- A commercial LLM key is available by Day 1 (Gemini or GPT-4o per the Module Spec).
- A Frappe Cloud site with custom-app support is reachable on Day 1 (R-14). Spike S1 validates this.
- One pilot studio with seeded demo data; two tenants seeded for isolation tests.
- Defaults are capacity 30, 4-hour cancellation cutoff, reminders at 12h/1h and 7/3/1 days.
- The OTP provider is unselected (spike S7); the fallback is a **mock OTP** in Beta.

Delivery constraints

- 6 people × 5 days (about 30 person-days). The scope is vertical slices, not full breadth.
- Two data stores (ERP and PostgreSQL) with explicit ownership.
- Frappe Cloud limits OS-level customisation, and the API-to-ERP hop is an added latency and coordination risk (R-03).
- The MVP still needs full-quality guards, audit and tenant scoping. These are never cut.

2. MVP Implementation Scope

Legend. P0 = mandatory in 5 days (Module Spec §10.2). P1 = important, built if P0 is green (Module Spec §10.3, with the PRD cut order). P2 = future.

2.1 B2B modules

Module	P0 (must ship)	P1	P2
Auth & RBAC	Staff login, forgot password, accept invite, JWT, route guards	Granular grants	MFA
Dashboard	Active members, sessions today, schedule with occupancy, 7-day renewals, quick actions, revenue tiles	Expiring-membership tile parity with Assistant	Trends, multi-studio
Members	Full CRUD, search, duplicate modal, archive	–	Bulk import, history tabs
Memberships	Plans, assign, approval queue, expiry, manual renewal, renewal pipeline	–	Freeze, credits
Classes	Class types, weekly schedule, single and recurring sessions, cancel session, roster	–	Blackout, substitution
Bookings	On-behalf, cancel, waitlist view	Auto-promotion message	Seat recovery
Attendance	Check-in dashboard, marking sheet, QR display, audited correction	–	No-show analytics
CRM	Lead capture (public form), status, conversion	–	Sequences, scoring
Events	Categories, create, registration management, free and paid	–	Event waitlist
Payments	List, manual record, Razorpay test, verified webhook	–	Refunds, reconciliation
Reports	Backed by dashboard services	Six core reports hub	Exports
Staff	List, invite, role assignment, deactivate	Permission grants	Payroll
Settings	Hub, studio profile, plans, class types, feature flags, booking rules	Notification templates	Integrations UI
Tenant mgmt	Seeded tenant → studio → branch → Owner	–	Self-service billing
Notifications (B2B)	Admin Notification Log (basic)	Full delivery-status log	Retry console

2.2 B2C modules

Module	P0	P1	P2
Auth	Phone OTP, Google OAuth, account link, registration	–	–
Dashboard	Status card, next 3 bookings, renewal CTA	Recommendation panel	Personalised highlights
Membership	Browse plans, checkout, confirmation, history	–	Freeze request
Bookings	Live schedule with eligibility reasons, book, cancel, waitlist	–	Conversational booking
Attendance	QR scan, success screen, history	–	Streaks
Events	Discover, register, pay	–	Waitlist
Payments	History	–	Receipts
Profile	View, edit	–	Preferences
Public site	Classes/events pages, enquiry form, workshop registration	Enquiry chat widget	Trial booking

2.3 ERP features

Priority	Items
P0	Custom app `wellness_studio` on Frappe Cloud; DocTypes (20 per the DB doc, fewer if a slice slips); whitelisted atomic methods (book, waitlist promote, check-in, membership activate); tenant permission hooks; expiry scheduler (idempotent); signed callbacks to the API
P1	Report aggregates for the six reports
P2	GST invoicing, accounting, reconciliation

2.4 AI features

Priority	Items
P1 (Day 4–5, flag-gated)	Admin Assistant: expiring-this-week, low-attendance, Create Member with propose/confirm and AI audit
P1 (cut order 1–3)	Booking suggestions, Recommendations, Feedback, Enquiry assistants (narrow)
P2	Conversational booking, approval queue, CRM and Operations agents

The AI foundations (LLM gateway, tool registry, AI audit, pending-action store) are built early and are the priority. If P0 slips, they are cut first.

2.5 Automation features (n8n)

Priority	Items
P0	Outbox dispatcher and HMAC signing; `send-message` sub-workflow (mock); Lead confirmation; Booking confirmation; Payment received;

	Event registration; Renewal reminders; Class reminders; Failure alert
P2	Event reminders 24h/6h/1h, inactive nudges, email

2.6 WhatsApp features

Priority	Items
P0	Mock mode with a would-send record (recipient, template, time); idempotency key (event ID + recipient + template); status stored per message
P1	Live Cloud API switch (config only)
P2	Templates UI depth, opt-in preferences, email

2.7 What MUST be done in 5 days

Tenant/studio/branch model; six-role RBAC; audit; Members; Memberships; Classes; Bookings (with waitlist); Attendance; Events (basic); Payments (test); CRM capture to conversion; Config basics; core dashboard; B2C portal P0 screens; outbox → n8n → WhatsApp mock for the core flows; and the E2E canonical-flow test with tenant-isolation tests on Beta.

Cut order if time slips (PRD §16): (1) Booking and Recommendation assistants; (2) Feedback; (3) Enquiry; (4) Paid-event payment; (5) Create Member action; (6) read queries. **Never cut:** core modules, permissions, tenant isolation, audit.

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3. Team Structure

Roles follow the ownership proposal in Architecture §14.7. Team members are referred to by role code throughout (SL, DBE, AIP, AIQ, ERPE, FEL); the team lead maps each code to a person before Day 1. The load is balanced by giving the two heaviest tracks (ERP and Frontend) a dedicated person each. The AI members carry Day 1–2 platform work plus Day 5 QA, so they are not idle while their tools depend on core modules.

Member	Role	Responsibilities	Modules owned	Deliverables
SL	Solution Lead / Backend Lead / TPM	Contracts (OpenAPI), auth, tenancy, RBAC, audit, outbox, ERP adapter, daily integration, PR gatekeeping, release	Auth, Tenants, Users/Roles, Audit, ERP adapter, Staff, Settings API	Contract freeze (D1), guards and audit (D1–2), adapter and outbox (D2–3), release tags
DBE	Domain Backend Lead + Automation	Domain services, API tests, n8n workflows, dispatcher worker, WhatsApp mock, CI/CD with SL	Members, Memberships, Classes, Bookings, Attendance APIs; `automation/n8n`; `apps/worker`	Domain APIs (D2–3), workflows and mock (D3–4), CI pipeline (D1)
AIP	AI Platform Engineer	LLM gateway, tool registry, guardrails, narrow assistants, evaluation harness	AI gateway/tools/guardrails; Enquiry, Booking, Recommendation, Feedback	Gateway (D2), tool defs (D1), read tools (D3), narrow assistants (D5)
AIQ	AI Assistant + QA Lead	Admin Assistant orchestrator, Create Member flow, pending-action store, chat panel, test plan and evidence	Admin Assistant; QA coordination; test docs	Loop (D2), Create Member (D4), evidence pack (D5)
ERPE	ERP / Frappe Engineer	Frappe Cloud bench, custom app, DocTypes, atomic methods, tenant hooks, scheduler, callbacks	`erp/wellness_studio`	Custom app on dev (D1), DocTypes (D1–2), atomic booking (D2–3), expiry job (D3)
FEL	Frontend Lead	Next.js B2B and B2C portals, design system, tenant theming, all UI states, Razorpay checkout	`apps/web`, `packages/ui`; Events/Payments/Reports UI	Shell and auth (D1–2), staff screens (D2–3), member portal (D3–4), polish (D5)

Buddy pairs (bus-factor cover): SL↔DBE (API/ERP/CI); AIP↔AIQ (AI); ERPE↔SL (ERP adapter); FEL↔AIQ (frontend QA and chat panel). Frontend is the tightest track. From Day 3, AIQ and AIP each take a defined set of screens once their AI foundations are stable (specifically the chat panel, recommendation panel, and enquiry widget).

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4. Work Breakdown Structure

IDs: FE = Frontend, BE = Backend, DB = Database, ERP, AI, AUT = Automation, TST = Testing, DEP = Deployment. Owner codes: SL Solution Lead, DBE Domain Backend/Automation, AIP AI Platform, AIQ AI Assistant/QA, ERPE ERP Engineer, FEL Frontend Lead.

Database (PostgreSQL app DB)

ID	Description	Owner	Depends on	Output
DB-1	Prisma schema and migrations for the P0 tables only (tenants, studios, branches, users, roles, permissions, config, audit, outbox, notifications, AI audit/pending actions, feedback)	SL	Contract freeze	Migrations run in dev and beta
DB-2	Row-level security and tenant session setting	T	DB-1	RLS tests green
DB-	Seed script: 2 tenants, roles, plans, class types, users, demo data	DBE	DB-1,	`pnpm seed`

3			ERP-2	idempotent
DB-4	Append-only audit permission	SL	DB-1	App role cannot update/delete audit

ERP

ID	Description	Owner	Depends on	Output
ERP-1	Frappe Cloud bench, custom app repo installed from GitHub (spike S1)	ERPE	–	App deploys from branch
ERP-2	DocTypes with tenant link	ERPE	Contract freeze	DocTypes deployed on dev
ERP-3	Tenant permission hooks (spike S2)	ERPE	ERP-2	Two-tenant denial test
ERP-4	Atomic `book`, `cancel`, `waitlist_promote` with row lock (spike S3)	ERPE	ERP-2	50-parallel test: 1 success
ERP-5	Membership activate/approve/expire and idempotent scheduler	ERPE	ERP-2	State machine enforced
ERP-6	Check-in and correction methods (immutable history)	ERPE	ERP-2	Duplicate rejected
ERP-7	Signed callbacks to API	ERPE + SL	BE-2	Callback accepted and audited
ERP-8	Payment posting method	ERPE	ERP-2	Payment record written

Backend

ID	Description	Owner	Depends on	Output
BE-1	Monorepo, NestJS skeleton, common (guards, interceptors, correlation ID, error codes)	SL	–	Skeleton running
BE-2	Auth (staff JWT, invite, reset; member OTP + Google), tenant guard, policy service	SL	BE-1, DB-1	Auth endpoints live
BE-3	ERP adapter with timeouts, idempotency keys, error mapping	SL	ERP-1	Health and version check
BE-4	Audit interceptor and outbox writer (same transaction)	SL	DB-1	Audit and outbox rows on writes
BE-5	Members (duplicate check, archive)	DBE	BE-3, ERP-2	CRUD + 409 on duplicate
BE-6	Memberships and plans	DBE	BE-5, ERP-5	Assign, approve, expire
BE-7	Classes and sessions (recurrence expander)	DBE	ERP-2	Sessions, capacity read from config
BE-8	Bookings and waitlist	DBE	ERP-4, BE-6, BE-7	Book/cancel/promote
BE-9	Attendance (QR, trainer, front desk, correction)	DBE	ERP-6, BE-8	Endpoints tested
BE-10	Events and registration (payment-hold timer)	ERPE → DBE	BE-11	Free and paid registration
BE-11	Payments: Razorpay order, verified webhook, manual record, ERP posting	SL	ERP-8	Idempotent webhook
BE-12	CRM (public form, conversion)	DBE	BE-5	Lead → member
BE-13	Dashboard/Reporting service, shared with the Assistant	SL	BE-5–BE-11	KPI payload
BE-14	Staff, Settings/config service + cache	SL	BE-2	Config effective without deploy
BE-15	Notification module: outbox dispatcher, log, what-is-due query	DBE	BE-4	Events reach n8n
BE-16	Internal automation API (n8n → API)	DBE	BE-15	Authenticated service credentials

Frontend

ID	Description	Owner	Depends on	Output
FE-1	Next.js shell, route groups, tenant theme, middleware host→tenant, API client generated from OpenAPI	FEL	Contract freeze	Shell + themed layout
FE-2	Design system (`packages/ui`) with four UI states	FEL	FE-1	Reusable components
FE-3	Staff login, invite, reset	FEL	BE-2	Login flow
FE-4	B2B: Dashboard, Members, Memberships (plans, assign, approvals)	FEL	BE-5, BE-6	Screens
FE-5	B2B: Classes, Session detail, Booking on-behalf, Attendance, QR display	FEL	BE-7–9	Screens
FE-6	B2B: Payments, Events, Staff, Settings, Notification Log	FEL (AIQ assist)	BE-10, 11, 14, 15	Screens
FE-7	B2C: OTP/Google login, dashboard, plans, checkout (Razorpay)	FEL	BE-2, BE-11	Purchase flow

FE-8	B2C: schedule/booking, QR scan, history, events, profile	FEL (AIP assist)	BE-8, BE-9	Screens
FE-9	Public pages (SSR) and enquiry form	FEL	BE-12	Public site
FE-10	Staff AI chat panel with confirmation card	AIQ	AI-6	Chat panel

AI

ID	Description	Owner	Depends on	Output
AI-1	Tool definitions and guardrail/prompt spec (D1)	AIP	Contract freeze	`packages/ai-tools`
AI-2	LLM gateway (provider-agnostic, timeouts, one retry)	AIP	AI-1	Tool-calling round trip (S6)
AI-3	Tool registry, permission filter, AI audit	AIP	BE-4	Audit on every call
AI-4	Orchestrator loop + pending-action store	AIQ	AI-2, AI-3	Propose → confirm state machine
AI-5	Read tools: expiring this week, low attendance	AIP	BE-13	Matches dashboard
AI-6	Create Member flow	AIQ	AI-4, BE-5	Missing fields, duplicate, permission, failure paths
AI-7	Enquiry, Booking, Recommendation, Feedback (narrow)	AIP	BE-7, BE-8, BE-12	Four thin endpoints
AI-8	Evaluation harness (curated queries, Create Member scenarios)	AIQ	AI-5, AI-6	Recorded results

Automation

ID	Description	Owner	Depends on	Output
AUT-1	n8n on Oracle VM, credentials template, import scripts	DBE	–	n8n reachable privately
AUT-2	`send-message` sub-workflow: template render, consent check, mock/live switch, retry, status	DBE	AUT-1, BE-16	Would-send record
AUT-3	Lead and Booking confirmation	DBE	AUT-2	Verified in log
AUT-4	Payment received, Event registration	DBE	AUT-2, BE-11	Verified
AUT-5	Renewal and class reminders (scheduled "what is due")	DBE	AUT-2, BE-15	No duplicates
AUT-6	Failure alert	DBE	AUT-2	Admin visibility
AUT-7	Dedup and outbox replay test (spike S5)	DBE	AUT-2	Duplicate event → one message

Testing

ID	Description	Owner	Depends on	Output
TST-1	Unit tests per module (rules, policies)	Each owner	Module code	CI green
TST-2	Concurrency test (50 parallel last seat)	ERPE + AIQ	ERP-4	Evidence
TST-3	Tenant isolation suite	SL + AIQ	ERP-3, DB-2	CI job
TST-4	Payment webhook tests (bad signature, duplicate, delayed)	SL	BE-11	Evidence
TST-5	Mock notification inspection	DBE	AUT-2–5	Evidence
TST-6	E2E canonical flow (Playwright or scripted)	AIQ + FEL	Integrations	Repeatable script
TST-7	AI evaluation run	AIQ	AI-8	Results recorded
TST-8	Smoke and regression checklists	AIQ	DEP-3	Signed checklist

Deployment

ID	Description	Owner	Depends on	Output
DEP-1	GitHub repo, branch protection, Actions (lint, type-check, unit, secret scan)	DBE	–	CI on PRs
DEP-2	Oracle VM with Docker Compose, reverse proxy, TLS/wildcard	SL + DBE	–	Beta URL
DEP-3	Deploy pipeline: ERP app → migrations → API/worker → web → n8n import	DBE	DEP-2	One-command deploy
DEP-4	Secrets injection, backups (PostgreSQL daily, n8n), rollback tags	SL	DEP-2	Restore tested once

DEP-4	Secrets injection, backups (`pg_dump` daily, n8n), rollback tags	SL	DEP-2	Restore tested once
DEP-5	Demo seed and reset script	DBE	DB-3	Repeatable demo state

5. 5-Day Execution Plan

Rhythm: 09:30 stand-up (10 min); 13:30 integration sync; 18:00 checkpoint and demo of the day's slice; merge freeze at 19:00 for that day's integration branch. Blockers are escalated to SL within 30 minutes.

Day 1 — Setup, Alignment, Database, Initialisation

Task	Owner
Resolve the source conflicts table; freeze P0 scope and OpenAPI contract for members, memberships, classes, bookings, attendance, leads, payments	SL (all attend, 90 min)
Monorepo, CI, branch protection, Docker Compose for local	DBE
NestJS skeleton, guards, correlation ID, tenant context	SL
Prisma migrations for P0 tables; RLS	SL
Spike S1 Frappe Cloud custom app from private repo; **S2** tenant isolation approach	ERPE
DocTypes (draft) for members, plans, memberships, classes/sessions, bookings, attendance, payments, events, leads	ERPE
Next.js shell, theme tokens, generated API client, design system base	FEL
Tool definitions, guardrail spec, FAQ source list; LLM key check	AIP
Test plan skeleton; mock-OTP decision (S7); demo script draft	AIQ
Oracle VM, Docker, n8n instance	DBE (with SL)

Deliverables. Frozen contract; repo and CI green; DB migrations applied to dev; ERP app installed on dev; UI shell rendering tenant theme; AI tool spec. **Checkpoint (18:00).** S1 and S2 outcomes known. If S1 fails, escalate immediately to the mentor. Fallback: install custom app via Frappe Cloud support or a self-hosted bench on Oracle; timebox 2 hours.

Day 2 — Core Module Development

Task	Owner
Auth (staff JWT/invite/reset), RBAC policy service, audit + outbox	SL
ERP adapter (health, version check, error mapping)	SL
DocTypes finalised; membership state machine; tenant hooks; **S3** atomic booking	ERPE
Members API with duplicate control; Plans and Memberships API	DBE
Login, Members list/form, Plans screens	FEL
LLM gateway and **S6** tool-calling round trip	AIP
Orchestrator skeleton and pending-action store	AIQ
Configuration Center API basics (plans, class types, capacity, cancellation window)	SL

Deliverables. Staff can log in; create members with duplicate rejection; create plans; assign a membership; audit rows written; S3 result. **Checkpoint.** Vertical slice 1: login → create member → assign membership passes via UI on dev. Tenant guard rejects cross-tenant call.

Day 3 — Integration Development

Task	Owner
Classes and sessions API; Bookings and waitlist; Attendance API	DBE
Atomic booking, cancel/promote, check-in methods, expiry scheduler	ERPE
Payments: Razorpay order, verified webhook (idempotent), manual record, ERP posting	SL
Outbox dispatcher, notification log, internal automation API	DBE (with SL)
Class/Booking/Attendance screens (B2B); OTP login; B2C dashboard and plans	FEL
Read tools (expiring, low attendance) wired to the Reporting service	AIP
Dashboard/Reporting service	SL
Tenant-isolation suite starts; concurrency test harness	AIQ

Deliverables. Canonical flow through the API: membership → session → booking → attendance → payment (Razorpay test). **Checkpoint.** API-level E2E script runs green on dev. Concurrency test: exactly one confirmation. **Scope gate:** if bookings or payments are not working, freeze all P1 work and swarm.

Day 4 — AI, Automation, Notifications (and Portals)

Task	Owner
n8n: `send-message` (mock), Lead, Booking, Payment, Event, Renewal, Class reminders, Failure alert; **S5**	DBE

dedup	
Create Member flow, confirmation card, true-result reporting, AI audit	AIQ
Events and registration API (paid hold); CRM public form and conversion	ERPE → DBE/SL split
B2C booking, QR, events, payments, profile; B2B payments/events/staff/settings screens	FEL (AIP helps with B2C profile and events)
Assistant chat panel	AIQ
Booking suggestion and Recommendation (narrow, rules first)	AIP
First deploy to Beta (through the pipeline)	SL + DBE

Deliverables. Mock notifications appear in the Notification Log for lead, booking, payment, event and renewal; Create Member works through chat; portals functional on Beta. **Checkpoint.** Deploy to Beta done by 18:00. Apply cut order 1 if the AI or portals are behind.

Day 5 — Testing, Bug Fixes, Deployment, Demo

Task	Owner
Fix-only mode from 12:00. **Feature freeze at 11:00.**	All
E2E canonical flow on Beta (TST-6); regression and smoke checklists	AIQ + FEL
Tenant isolation, role-scope API tests, webhook tests	SL + AIQ
Enquiry and Feedback assistants (only if the freeze allows); AI evaluation run and results	AIP + AIQ
Notification verification; demo seed and reset	DBE
Final release tag, backup, rollback tag recorded	SL
Demo rehearsal ×2 (14:00 and 16:00); recording as fallback	All

Deliverables. Tagged release on Beta; evidence pack (test results, AI evaluation, isolation and concurrency results); demo script and seed. **Checkpoint.** Go/no-go at 15:00 against the success-criteria table. Sev-1 open = no-go for that item; demo around it.

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6. Git & Branch Strategy

Aspect	Rule
Repository	One private GitHub monorepo `wellness-studio-os` (`apps/*`, `packages/*`, `erp/`, `automation/`, `infrastructure/`, `docs/`)
Long-lived branches	`main` (tagged releases), `develop` (integration, auto-deploys to Dev), `release` (Beta)
Naming	`feature/<module>-<short-desc>` (e.g. `feature/bookings-waitlist`), `fix/<id>-<desc>`, `chore/<desc>`, `spike/<S#>-<desc>`
Feature branches	Small and vertical, less than about 4 hours old before merge. Cut from `develop`. Ownership follows repo areas in §3.
Pull requests	Template with description, contract change yes/no, tests, screenshots. CI must pass (lint, type-check, unit, secret scan). One review required; **SL reviews all contract, guard, audit and adapter changes.** Review response target under 1 hour.
Merge strategy	Squash-merge into `develop`. Fast-forward `develop` → `release` at day-end after the checkpoint. Tag `v0.1.0` on `main` for the demo release.
Contract changes	OpenAPI updated in `packages/contracts` **first**, in its own PR, announced in stand-up. The Frappe app deploys before the adapter.
Frappe app	Own path `erp/wellness_studio`. Merges to `develop` update the dev site's custom-app branch.
Hard rules	No secrets, `node_modules`, venvs or DB dumps in Git. No direct pushes to `develop`, `release` or `main`.
Conflict resolution	(1) Rebase your branch on `develop` daily. (2) The author of the later PR resolves conflicts. (3) For shared files (`contracts`, `common`, migrations) the owner (SL) arbitrates in a 10-minute call. (4) Migration conflicts: renumber to the next timestamp; never edit a merged migration. (5) If unresolved in 30 minutes, the lower-priority feature waits.

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7. Integration Plan

Sequence (each link has a smoke check before the next one starts):

Order	Link	How	Smoke check	Owners	Day
1	Backend ↔ Database	Prisma migrations, RLS, tenant context	Insert/select under two tenants	SL	D1–2
2	Backend ↔ ERP	Adapter to whitelisted methods; restricted API identity; timeouts; idempotency key on writes; startup version check	Create member through the API appears in ERP; cross-tenant denied	SL + ERPE	D2
3	Frontend ↔ Backend	Typed client generated from OpenAPI; httpOnly cookies; host→tenant	Login → member list on dev	FEL + SL	D2–3
4	Backend ↔ Razorpay	Order creation → client checkout → signed webhook → payment Paid → ERP posting → outbox `payment.confirmed`	Test payment; replayed webhook creates one record; bad signature	SL + FEL	D3

			rejected		
5	Backend ↔ n8n	Outbox dispatcher → HMAC webhook; n8n → internal automation API with service credentials	`booking.confirmed` arrives once; duplicate delivery ignored (S5)	DBE	D3–4
6	Backend ↔ WhatsApp (via n8n)	Mock switch inside `send-message`; would-send record; status callback stub	Log record has recipient, template, time	DBE	D4
7	Backend ↔ AI	Gateway → tool registry → same domain services; pending-action store; AI audit	Read query equals dashboard; Create Member propose → confirm	AIP + AIQ	D3–4

Critical dependencies (critical path): Contract freeze → ERP-1/S1 → DocTypes + tenant hooks → adapter → domain APIs → atomic booking → payments → outbox → n8n → mock messaging → UI → E2E. Second path: dashboard/Reporting service → AI read tools → Create Member.

Dependency	Why it matters	Trigger and fallback
Frappe custom app installable (S1)	Everything downstream	Not resolved by Day 1 noon → fallback bench on Oracle
ERP tenant approach (S2)	Isolation is a never-cut item	Company + permission hooks; if it fails, site-per-tenant for the two demo tenants
Atomic booking (S3)	Core acceptance gate	Row lock in the custom-app method; no application-level workaround
Outbox after ERP success	Commit-before-notify	Write the outbox only after the ERP call succeeds
LLM key and rate limits	AI slice	Switch provider via gateway config
OTP provider (S7)	B2C login	Mock OTP on Beta

8. Testing Plan

Level	Scope	Owner	When
Unit	Policies, rule functions (eligibility, capacity config, cutoff, state machines), DTO validation, audit writer, outbox, AI tool wrappers, gateway retries; Frappe rule tests in `erp/.../tests`	Each module owner	Continuous, in CI
Integration	API ↔ ERP adapter; API ↔ DB (RLS); Razorpay webhook (signature, duplicate, delayed); outbox → n8n → mock log; ERP callbacks; AI tools → services; concurrency (50 parallel)	SL, DBE, ERPE, AIP	D2–D4
Tenant isolation	Every P0 module, two tenants; 403/not-found style response, audit entry	SL + AIQ	D3 start, CI from D4
AI evaluation	Curated query set for the two read queries; Create Member scenarios (happy, missing fields, duplicate, permission denied, API failure); targets 85% / 95% / zero false success	AIQ + AIP	D4–D5
Smoke (after each deploy)	Login (staff, member) → create member → book → payment test → notification log entry → dashboard loads	AIQ (DBE runs the script)	Every Beta deploy
Regression	Full canonical flow, role matrix (Owner/Admin/Front Desk/Trainer/Member), booking boundary (membership expiry), duplicate detection, attendance correction audit, settings change without deploy	AIQ + FEL	D5 morning, again after fixes
UAT	Demo-journey walkthrough per persona (owner, front desk, trainer, member, visitor), executed by someone who did not build the feature (pair swap)	Pair swap, coordinated by AIQ	D5 13:00

Execution order: unit (author, before PR) → integration (on `develop`) → smoke (each deploy) → isolation + concurrency + webhook (D3–D4) → AI eval (D4–D5) → full regression (D5) → UAT → go/no-go.

Defect policy: Sev-1 blocks the release; Sev-2 is fixed if under 1 hour, otherwise logged and waived at go/no-go; Sev-3 is logged for Phase 2. Every fix is reviewed and re-smoked.

Evidence: results are stored in `docs/test-evidence/` (required by the PRD's Definition of Done).

9. Deployment Plan

Environments (Phase 1 per Architecture §13.1): Development (local Docker Compose plus shared dev host on Oracle, shared Frappe Cloud dev site, shared dev n8n) and **one Beta/Demo environment** (Oracle VM containers, separate PostgreSQL seeded with demo data, Frappe Cloud staging site, staging n8n). Production is Phase 2.

Environment setup (Day 1–4)

- Oracle VM with Docker Compose: Next.js, NestJS API, worker (same image), n8n, PostgreSQL, reverse proxy with wildcard TLS. Only 80/443 public.
- Environment variables per environment: ERP URL and API identity, Razorpay test keys and webhook secret, HMAC secret for n8n and ERP callbacks, JWT secrets, LLM key, mock/live flags for WhatsApp and Razorpay (mock/test), OTP mode.
- Wildcard DNS for tenant subdomains (e.g. `pilot.<domain>`).
- Frappe Cloud staging site with the custom app deployed from `release`.

Deployment sequence (matches Architecture §13.3)

1. Merge to `release` and pass CI.
2. Deploy the **Frappe custom app**.
3. Run PostgreSQL **migrations** (backward-compatible only).
4. Deploy **API and worker**.
5. Deploy **web**.
6. **Import n8n workflows** from the repo.
7. Run the **seed** (demo tenant/users/data) and smoke script.
8. Watch logs for 15 minutes.

Verification checklist: health endpoints (API, worker, ERP version check); staff and member login; create member; book with the same-day capacity check; Razorpay test payment produces Paid state; one Notification Log record appears in mock mode; the dashboard shows the seeded numbers; a cross-tenant call is denied; the Assistant answers one read query.

Rollback considerations

- Tag every release. Roll back API/web by redeploying the previous container tags.
- Roll back the Frappe app via Frappe Cloud app rollback. The adapter's startup version check fails fast on a mismatch.
- Use expand → migrate → contract migrations, so no destructive change ships in the 5-day window.
- Take a PostgreSQL backup and a Frappe Cloud backup before the D5 release; test one restore on D4/D5.
- n8n: re-import the previous workflow export. The outbox retains undelivered events, so replay is safe because consumers are idempotent.
- Feature flags disable AI and unfinished capabilities without redeploying.

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10. Risk Management

I = Impact, P = Probability (H/M/L).

Category	Risk	I	P	Mitigation
Technical	5-day scope too large (R-01)	H	H	Strict P0 list; vertical slices; documented cut order; Day 3 scope gate; feature freeze D5 11:00
Technical	Contract drift between FE/BE/ERP	M	M	Contract-first OpenAPI; generated client; contract PRs reviewed by SL
Technical	Two data stores drift (ERP vs PostgreSQL)	M	M	Reference by ID only; ownership table; nightly reconciliation deferred to Phase 2, manual checks on D5
Technical	Booking p95 exceeds target due to the extra hop (R-03)	M	M	Thin adapter; S4 measurement on D3; skip caching for availability
Integration	Razorpay webhook duplicated, delayed or forged	H	M	Signature verification, idempotent by provider ID, Pending on delay; tests TST-4
Integration	n8n duplicate messages or outage	M	M	Idempotency key; outbox queues during outage; S5 test
Integration	ERP unavailable during demo	H	L	Explicit "temporarily unavailable" behaviour; keep a recorded demo as fallback
Integration	Frappe Cloud rejects or restricts the custom app (R-14)	H	L–M	S1 on Day 1; fallback bench on Oracle
AI	Wrong or unintended action (R-05)	H	L	Propose → confirm; stored arguments only; permission inheritance; audit
AI	Inaccurate answers or invented data (R-06)	M	M	Answers only from tool results; scope refusal; curated queries
AI	LLM latency or quota	M	M	Timeouts; one retry; gateway provider switch; "assistant unavailable" degrade path
AI	AI work crowds out core work	H	M	AI last in the sequence; first in the cut order; AI members help on core screens on D3
ERP	Learning curve and DocType complexity (R-02)	H	H	Dedicated ERPE; buddy SL; DocTypes limited to P0 slices; time-boxed spikes
ERP	Tenant isolation not achievable in the ERP	H	L–M	S2 on Day 1; site-per-tenant fallback for two demo tenants
ERP	Scheduler jobs run twice	M	L	Idempotent jobs by design and test
Team	Single points of failure (ERPE on ERP, FEL on frontend)	H	M	Buddy pairs; daily code walkthrough; AI members take defined screens from D3
Team	Uneven experience (R-13)	M	M	Pair on the first vertical slice; common Definition of Done
Team	Role-to-person mapping not confirmed	M	M	Team lead confirms role assignments before Day 1 stand-up
Timeline	Late credentials (Razorpay test, LLM, OTP, WhatsApp)	M	M	Mock modes; credentials checked Day 0; mock OTP fallback
Timeline	Merge conflicts and late integration	M	M	Daily integration branch; day-end freeze; small PRs
Timeline	Demo-day surprise defects	H	M	Feature freeze D5 11:00; two rehearsals; recorded fallback; seed reset script

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11. MVP Demo Plan

Target length is about 20 minutes on the Beta environment, with pre-seeded data (two tenants, so the isolation moment is real) and a scripted reset.

#	Journey	Steps	Proves
1	**Studio Owner**	Log in (pilot tenant, branded theme) → dashboard tiles → create a "10-class pass" plan in Settings (no deploy) → add a Pilates class type → publish a paid workshop with 20 seats → show the audit entry	Configuration without developers; tenant branding; audit
2	**Staff (Front Desk, Trainer)**	Front Desk registers a walk-in (duplicate phone → warning modal) → assigns a membership → approval queue (Admin approves) → books on behalf → last-seat contention run shown → Trainer sees only assigned sessions, no revenue → marks attendance → Admin corrects one record with a reason	Duplicate control; eligibility; capacity; RBAC; audit
3	**Member**	Phone OTP login → dashboard → browse plans → checkout (Razorpay test) → membership active → schedule shows eligibility reasons → book, join waitlist, cancel inside cutoff (rejected with reason) → QR check-in → history	B2C self-service; booking rules; QR
4	**Payment**	Successful test payment → status Paid only after webhook → replay the webhook (no duplicate) → a pending/failed payment displayed as not paid	Verified confirmation; idempotency
5	**WhatsApp (mock)**	Show the Notification Log: lead, booking, payment, event and renewal entries with recipient, template, time, delivery status → show one failed message surfaced to admin without reversing the booking	Outbox, mock mode, failure visibility
6	**AI**	Ask "memberships expiring this week?" → answer equals the dashboard tile → ask "classes with low attendance" → "Add Rahul, Premium, starting Monday" → assistant asks for missing phone → confirmation card with the resolved date → confirm → real record ID → try again to trigger a duplicate error, reported truthfully → show the AI audit row; Front Desk user lacks the create tool	Governed AI, propose/confirm, no false success
7	**Visitor** (if time allows)	Public class page → enquiry form creates a lead → confirmation in the log → convert to member	CRM lead to member
8	**Close**	Cross-tenant attempt returns denied, with the audit record → state what is in P1 and P2	Isolation; roadmap

Roles in the demo: SL presents the architecture and isolation, FEL drives the B2B/B2C screens, AIQ runs the AI segment, DBE the WhatsApp and automation segment, ERPE the concurrency demo, and AIP handles Q&A on the AI features. Every segment has a pre-recorded fallback clip.

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12. Go-Live Checklist

Phase 1 goes live on the **Beta/Demo environment** (pilot studio). Items marked ★ are known Phase 2 gates before a true production release.

Frontend

- [ ] All P0 screens present; loading, empty, error and permission-denied states on each
- [ ] No console, type or lint errors; mobile viewport checked for trainer and member flows
- [ ] No optimistic updates for booking or payment; no secrets in client code
- [ ] Tenant theme loads from host

Backend

- [ ] JWT required on every protected route; tenant taken from the token and bound to the host
- [ ] Guards and policy service used everywhere (no inline role checks)
- [ ] Validation, rate limits, CORS allow-list, request-size limits on public, OTP and AI endpoints
- [ ] Idempotent create endpoints; errors normalised, no sensitive data in errors or logs
- [ ] Correlation ID flows UI → API → ERP → n8n

Database

- [ ] Migrations applied cleanly; RLS active and tested
- [ ] Audit table append-only for the app role
- [ ] Seed idempotent; demo data reset script works
- [ ] Daily backup configured and one restore verified

ERP

- [ ] Custom app deployed from `release`; version check passes
- [ ] DocTypes and whitelisted methods deployed; tenant permission hooks active
- [ ] Atomic booking verified (50 parallel → 1 success); scheduler idempotent
- [ ] Restricted integration identity (not administrator); ERP not exposed to browsers
- [ ] ★ GST invoicing and accounting configuration

AI

- [ ] AI features behind tier and feature flags
- [ ] Every AI call has an audit row; tools filtered per user and re-checked at execution
- [ ] Propose → confirm verified; no false-success cases in evaluation; results recorded
- [ ] LLM timeouts and "assistant unavailable" degradation verified; PII minimised in prompts and logs

WhatsApp

- [ ] Mock mode active; would-send records complete
- [ ] Approved-template list defined; consent check in the API
- [ ] Idempotency verified; failed sends visible to admins
- [ ] ★ Production credentials swapped in via configuration only; template approval

Automation

- [ ] All P0 workflows imported from the repo; signed webhooks verified (HMAC)
- [ ] Outbox backlog and failure alert observed
- [ ] Reminder schedules produce no duplicates

Security

- [ ] Secrets only in environment/secret store; none in Git (secret scan green)
- [ ] Razorpay and n8n signature verification tested with bad signatures
- [ ] Cross-tenant test suite passes; denials audit-logged
- [ ] Role-scope API tests pass (Trainer, Front Desk, Member)
- [ ] ★ MFA for Owner and Admin; legal review of consent and opt-in; security and performance review

Deployment

- [ ] HTTPS with wildcard cert; only 80/443 public
- [ ] Deployment sequence executed in order; smoke script green
- [ ] Release tag, previous-tag rollback path, and pre-release backups recorded
- [ ] Demo runs from the deployed environment; recorded fallback available
- [ ] Sev-1 count is zero; open Sev-2/3 recorded and accepted at go/no-go